TRR Sample Format Title

Author Name (Academic Example)
Department of XXX
Institution Name, City, State or Country, and Postcode
Email: author@university.edu

Author Name (Public Sector Example)
Position
XXStateXX Department of Transportation
Department (if applicable)
City, State or Country, Postcode
Email: abc@dot.gov

Author Name (Private Practitioner Example)
Position
Company
City, State or Country, Postcode
Email: enj@abc.com

Submitted [Submission Date]

ABSTRACT

The reliability of autonomous vehicles (AVs) depends heavily on robust perception systems capable of detecting hazards in complex urban environments. While modern object detection systems perform well on common road entities, they struggle with High-Stakes, Low-Frequency (HSLF) hazards such as fallen electric cables. This paper introduces the Urban Fallen Cable Detection Dataset (UFCDD), a region-specific benchmark collected in Mirpur, AJK, Pakistan. A customized YOLOv5-based framework optimized for thin-structure detection achieves 96% mAP@50 and 76% mAP@50–95 with 15.8 GFLOPs, demonstrating real-time feasibility for smart mobility applications.

Keywords: Autonomous Vehicles, Smart Mobility, Thin Object Detection, YOLOv5, Urban Hazard Detection, Dataset Benchmarking

INTRODUCTION

Autonomous vehicles (AVs) are considered a key advancement in modern transportation systems. They are classified into six levels of automation (SAE Levels 0–5) and are expected to improve mobility, efficiency, and road safety significantly. According to the World Health Organization, road traffic accidents cause more than 1.35 million deaths annually, with human error being the dominant factor. By reducing risks associated with fatigue, distraction, and impaired driving, AVs have the potential to substantially decrease accident rates. In addition, they contribute to environmental sustainability through optimized traffic flow and improved fuel efficiency.

A core component of autonomous driving is the perception system, responsible for detecting, classifying, and localizing surrounding objects. Modern perception architectures rely on multi-sensor fusion, integrating cameras, LiDAR, and radar. Cameras provide semantic and visual context, LiDAR supplies three-dimensional spatial information, and radar ensures robustness under adverse environmental conditions. Deep learning models, particularly Convolutional Neural Networks (CNNs) and hybrid architectures, enable automatic hierarchical feature extraction for object detection tasks. However, while these systems perform well in detecting structured road elements, they remain vulnerable to thin, irregular, and low-frequency hazards. One critical yet underexplored category of such hazards is fallen or low-hanging electrical cables.

These represent high-stakes, low-frequency (HSLF) threats. Failure to detect them may result in severe vehicle undercarriage damage, electrical short circuits, loss of control, and serious safety consequences [5]. The risk is particularly pronounced in unplanned and densely populated urban environments in developing countries, including Pakistani settings, where infrastructure variability and maintenance limitations pose operational challenges not sufficiently addressed in Western-centric standards [6]. Environmental factors such as extreme weather events and dust storms further increase the likelihood of cable-related incidents [7].

From a technical standpoint, fallen cables present intrinsic geometric and photometric challenges. Their extremely low aspect ratio and minimal pixel representation make them difficult to distinguish from background noise, shadows, or road textures. LiDAR often produces sparse point-cloud returns for thin structures, and radar cross-sections may be insufficient for reliable detection. Consequently, perception systems frequently produce high false-negative rates, creating a critical safety gap [7].

This limitation is further compounded by deficiencies in standard autonomous vehicle datasets. Widely used benchmarks such as KITTI, NuScenes, and BDD100K primarily focus on common traffic participants and structured road objects. Thin, irregular hazards such as fallen cables are largely underrepresented or absent [8]. As a result, deep-learning models lack sufficient training data to learn fine-grained curvilinear features, limiting their robustness in unstructured urban environments.

To address this gap, this study introduces the Urban Fallen Cable Detection Dataset (UFCDD), a specialized dataset consisting of over 2,500 annotated images collected primarily in Mirpur, AJK, Pakistan. The dataset captures diverse environmental and lighting conditions to improve model generalization and robustness [6]. By focusing on high-risk, infrastructure-related hazards often overlooked in conventional datasets, UFCDD contributes geographical and infrastructural diversity essential for global AV deployment [15].

Building on UFCDD, this research develops a specialized CNN-based object detection framework using the YOLOv5 architecture. YOLOv5 offers an effective trade-off between detection accuracy and computational efficiency, making it suitable for real-time, safety-critical applications [1, 7, 8, 9, 10]. The model is trained to classify cable states as Safe (intact) and Unsafe (fallen or hazardous).

The proposed framework is empirically validated using strict evaluation metrics. It achieves a mean Average Precision (mAP@50) of 96% and mAP@50–95 of 76%, demonstrating strong classification performance and precise localization of thin structures. The higher-threshold evaluation confirms robustness in detecting geometrically complex, low-pixel-count objects, which is critical for safe path planning and hazard avoidance [13, 17, 18].

Furthermore, this work addresses an important domain-transfer gap. Previous line-detection studies primarily focused on aerial inspection using UAVs under high-contrast conditions [10,11]. In

contrast, ground-based AV systems must operate in cluttered urban scenes with irregular cable geometries and dynamic backgrounds. By targeting infrastructure-specific hazards common in developing regions [6], this research supports scalable and globally adaptable autonomous mobility solutions.

In summary, this study contributes:

- A dedicated dataset (UFCDD) addressing overlooked HSLF cable hazards [14].
- A specialized YOLOv5-based detection framework optimized for thin-structure recognition [1, 7, 8, 9, 10].
- Quantitative validation demonstrating robust performance under strict localization metrics [13, 17, 18].

These contributions help bridge a critical safety gap in autonomous vehicle perception, particularly in complex and unstructured urban environments.

RELATED WORK

Achieving full autonomous mobility (SAE Levels 4 and 5) requires highly reliable perception systems capable of maintaining environmental awareness under diverse operational conditions [1]. In the autonomous driving pipeline, perception provides safety-critical input to downstream modules, including localization, planning, and control [2].

Modern AV perception relies on multi-modal sensor fusion to compensate for the limitations of individual sensors [3]. Cameras provide rich semantic and contextual information, such as color, texture, and lane markings, and are commonly integrated with deep learning detectors such as YOLO [42] and Faster R-CNN [13] for object recognition [1]. However, vision systems degrade significantly in adverse weather or illumination conditions [7].

LiDAR and radar complement vision-based sensing. LiDAR generates accurate three-dimensional (3D) point clouds for localization and obstacle avoidance [1], while radar provides reliable distance and velocity measurements and remains robust in rain, fog, and dust [4]. Despite this redundancy, thin and geometrically complex objects such as fallen cables remain difficult to detect. LiDAR produces sparse returns for low-aspect-ratio structures, and radar cross-sections are often insufficient [5]. Consequently, detection of thin hazards relies heavily on high-resolution vision models and specialized deep learning frameworks [5].

High-Stakes, Low-Frequency (HSLF) Hazards

While AV perception systems perform well on structured traffic participants, they struggle with uncommon and irregular hazards [1]. Fallen electrical and telecommunication cables represent a critical class of High-Stakes, Low-Frequency (HSLF) obstacles [1]. Collisions with such hazards may cause severe mechanical damage, electrical system failure, and loss of control [1]. Organizations such as AAA and FAA have highlighted risks associated with power-line-related incidents [17]. The challenge is amplified in informal and densely populated urban environments such as Mirpur, Pakistan, where infrastructure density, limited maintenance, and environmental stressors increase cable-fall likelihood [25]. Reliable detection of such geographically specific hazards is essential for scalable global deployment of autonomous systems [25].

Geometric Ambiguity and Domain Shift

Fallen cables pose intrinsic geometric and photometric challenges. Their extremely low aspect ratio results in minimal pixel representation, often only 1–3 pixels in high-resolution imagery [43]. This creates ambiguity with shadows, cracks, road markings, and background clutter [45]. Variations in illumination further reduce contrast, leading to domain shift when models trained on structured datasets are deployed in unstructured environments [45]. Such models frequently fail to generalize to rare HSLF hazards [45].

Prior work in thin-object and cable detection provides useful foundations. Studies on aerial power-line extraction [9], segmentation of overhead lines [36], synthetic wire detection [44], thin-structure obstacle detection [58], and boundary-aware learning [50] demonstrate that attention mechanisms and segmentation-based approaches improve thin-object sensitivity [51]. However, most approaches focus on aerial imagery or controlled environments, limiting

transferability to ground-based AV scenarios.

Early research on utility inspection relied on UAV imagery and classical techniques such as Hough and Radon transforms for line reconstruction, often combined with Kalman filtering for tracking [1, 48]. Although effective for structured overhead lines, these methods fail in cluttered ground environments and cannot handle curved, non-rigid fallen cables [1]. Sensitivity to shadows and road texture variations further limits robustness [1].

The limitations of handcrafted feature extraction motivated the transition to deep learning [54]. Convolutional Neural Networks (CNNs) automatically learn hierarchical representations and improve robustness under varying environmental conditions [1]. However, transferring aerial inspection techniques to ground-level AV detection remains challenging due to increased background clutter and lower contrast between cables and road surfaces [22].

Real-Time Detection Architectures

Autonomous driving requires low-latency object detection on edge hardware [1]. Modern detectors are categorized into two-stage (e.g., Faster R-CNN) and single-stage (e.g., YOLO, SSD) models [1]. Single-stage detectors, particularly YOLO-based frameworks, are widely adopted due to real-time inference capabilities. YOLOv5 balances speed and accuracy through a backbone–neck–head architecture with multi-scale feature integration [1].

To improve thin-object detection, lightweight backbones such as MobileNetV4 [5] reduce computational overhead, while high-capacity architectures like ConvNeXt enhance fine-grainedfeature extraction [2, 59]. Multi-scale feature fusion using Feature Pyramid Networks (FPNs) preserves low-level edge details critical for Small Object Detection (SOD) [4, 61]. Hybrid CNN–Transformer approaches further enhance contextual reasoning for elongated structures [55].

Thin and Curvilinear Structure Perception

Traditional bounding-box detectors struggle with objects occupying only a few pixels [23, 24]. Semantic segmentation provides pixel-level delineation and is more suitable for thin hazards [25, 26], with architectures such as FCNs and U-Net forming foundational models [27]. Alternative approaches, such as the key-point encoding model, curvilinear structures as discrete point sets for curve reconstruction [1].

Robust detection also requires depth validation to distinguish cables from shadows or cracks [46]. Joint detection and depth estimation frameworks improve 3D reasoning [59], and additional sensor modalities may enhance redundancy [60].

Data Scarcity and Class Imbalance

Thin-cable detection presents an Extreme Class Imbalance (EVI) problem, in which cable pixels are vastly outnumbered by background pixels [46]. Standard cross-entropy loss biases models toward predicting the background, increasing false negatives [66].

Specialized loss functions such as Focal Loss [48] reduce the dominance of easy negatives, while Dice and Tversky losses optimize spatial overlap for segmentation under imbalance [66]. Boundary-aware losses further improve contour precision [66].

Synthetic Data Generation (SDG) methods, including GANs and diffusion models, mitigate data scarcity for rare hazards [15, 26]. Pre-training on synthetic variations followed by fine-tuning on real-world datasets improves robustness [66]. However, domain-specific real imagery remains essential to preserve realism, as emphasized for UFCDD collected in Mirpur, AJK [15, 66].

Dataset Deficiencies and Domain Gaps

Major AV benchmarks such as KITTI, NuScenes, and BDD100K emphasize structured traffic scenarios [35]. Thin objects are often filtered out during annotation due to insufficient confirming sensor returns [44], creating a systemic blind spot for low-profile hazards [35].

Unstructured ground datasets like RUGD [50] address terrain complexity but lack cable-specific annotations. Road debris datasets such as HazardNet focus on larger obstacles and exclude low-height curved infrastructure elements [24]. No existing dataset provides both fine-grained cable annotations and realistic unstructured urban complexity.

Synthesis

This review identifies a critical safety gap in detecting fallen cables within unstructured urban environments [31]. The gap arises from:
Geometric constraints: thin structures evade LiDAR and radar detection [32].
Architectural trade-offs: balancing precision and real-time performance remains challenging [43].
Data deficiency: existing datasets systematically exclude HSLF hazards [46].

Bridging this gap requires specialized architectures with multi-scale fusion [66], advanced loss strategies to mitigate imbalance [48, 66], and domain-specific datasets such as UFCDD comprising over 2,500 annotated urban images [39, 51, 66].

Empirical evaluation of a YOLOv5 framework trained on this dataset achieved 96% mAP@50 and 76% mAP@50–95 [66], demonstrating that targeted data and architecture design can significantly improve the detection of thin, irregular cable hazards. Future work should integrate high-fidelity synthetic scaling and 3D depth validation to ensure robust deployment across diverse global urban environments [37, 45, 66].

DATASET: UFCDD & METHODOLOGY

Synthesis

Detecting fallen electric and communication cables is a high-stakes but low-frequency hazard. This requires a specialized method that addresses both the lack of data and the need for reliable algorithms. Our approach uses two main parts. First, we developed the Urban Fallen Cable Detection Dataset (UFCDD). This is a high-quality dataset that covers many different urban conditions. Second, we created a custom YOLOv5 framework. This system is optimized to detect thin and curved structures in busy environments in real time. The process shown in Image 3.1 turns raw sensor data into safety decisions for autonomous vehicles.

Data Acquisition and UFCDD

The UFCDD includes more than 2,500 images. We collected these using 1.5-meter and 2.7-meter poles to capture real-world cable hazards. We focused our data collection on Mirpur, AJK, Pakistan. This region has unstructured urban settings that are usually missing from standard vehicle testing. We gathered both still images and moving video to include different perspectives and background distractions.

Image Acquisition Platform

We used a GoPro Hero 8 camera for several reasons. It has a high frame rate of 120 fps and 10-bit color depth for sharp images. It also offers strong stabilization to reduce vibrations on bumpy roads. Additionally, it performs well in low light, which helps capture cable details during dawn or dusk. We mounted the camera on the vehicle's hood to mimic a car’s forward-facing vision system.

Environmental and Temporal Diversity

We collected images across different settings to make sure the model works well in general. This included busy streets and residential suburbs with varying infrastructure. We also captured data in sunny, cloudy, and rainy weather from morning until night. By combining static images with moving video, we were able to capture distortions caused by motion. This resulted in a benchmark dataset specifically for detecting fine-grained cable hazards.

Data Preparation and Ground Truth Annotation

Preprocessing and Key Frame Selection

We processed the videos to improve visual quality. We removed noise from low-light clips and adjusted the dynamic range. This made the cables easier to see against complex backgrounds. We then picked out key frames to avoid repetitive data while keeping a diverse range of images.

Manual Annotation

We labeled each frame by hand with high precision using the V7 Image Labeling Tool. We focused on two categories. The first is "Unsafe" for fallen or low-hanging cables. The second is "Safe" for fixed cables that are not a hazard. Accurate labeling helps the model tell the difference between thin cables and a cluttered background.

Dataset Partitioning

We split the UFCDD into three parts as shown in Table 3.1. The training set comprises 70% and is used for feature learning [28]. The validation set is 20% and is used to tune settings and prevent errors [58]. The final 10% is the testing set used to evaluate how the model handles new images [63].

Specialized Model Architecture

YOLOv5 Foundation

We chose YOLOv5 because it is fast and efficient. This is vital for safety in autonomous driving. The architecture has three main parts. The "Backbone" extracts spatial features. The "Neck" gathers features at different scales to help find small objects. Finally, the "Head" predicts whether a cable is safe or unsafe and identifies its location. Figure 3.1 shows this structure.

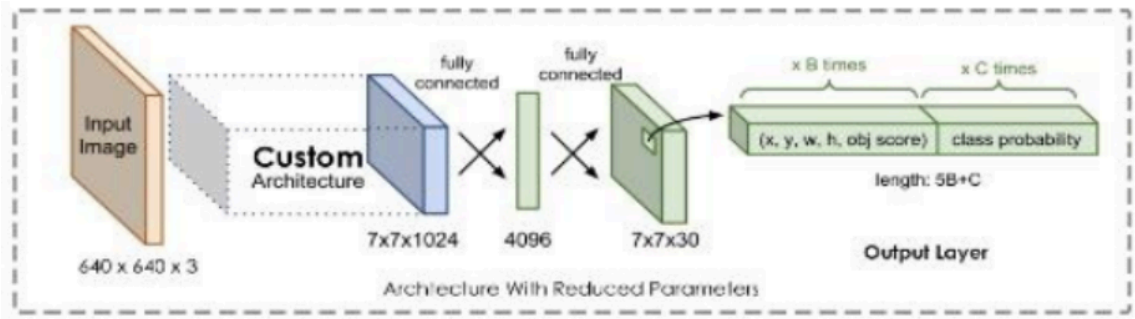

Figure 1 shows the YOLOv5 architecture.

Custom Configuration

We adjusted the settings to detect thin cables better. We set the width at 0.50 to balance power and efficiency. We set the depth to 0.33 to control the number of feature layers. The network uses special anchors to detect cables that are far away, small, or near the vehicle. This is essential for detecting small objects. You can see the performance details in Table 3.2.

Training Strategy

Computational Environment

We used an NVIDIA RTX 2080Ti GPU and an Intel Core i9 9900K CPU. The system had 32 GB of RAM. We ran the software using TensorFlow with OpenCV for faster processing.

Hyperparameters

We set the training for 300 cycles to ensure The model could recognize small hazards. We used a batch size of 32 for stability and set the input size to 640 pixels to capture detail without slowing down. We also used techniques like Label Smoothing and Early Stopping to keep the results consistent.

Table 1: summary of hyperparameters.

Hyperparameter	Value	Rationale
Epochs	300	Full convergence for low-pixel hazards
Batch Size	32	Stability and computational efficiency
Input Size	640 px	Detail capture without slowdown
Optimizer	SGD	Robust convergence

Loss Function

The YOLOv5 system uses a combined loss method. It uses "Bounding Box Loss" for precise cable locations and "Object Loss" to detect if a cable is present. It also uses "Classification Loss" for the safe and unsafe labels. In the future, we will add new methods to handle rare pixels and reduce missed detections. Figure 3.2 shows the full workflow from collecting data to making predictions.

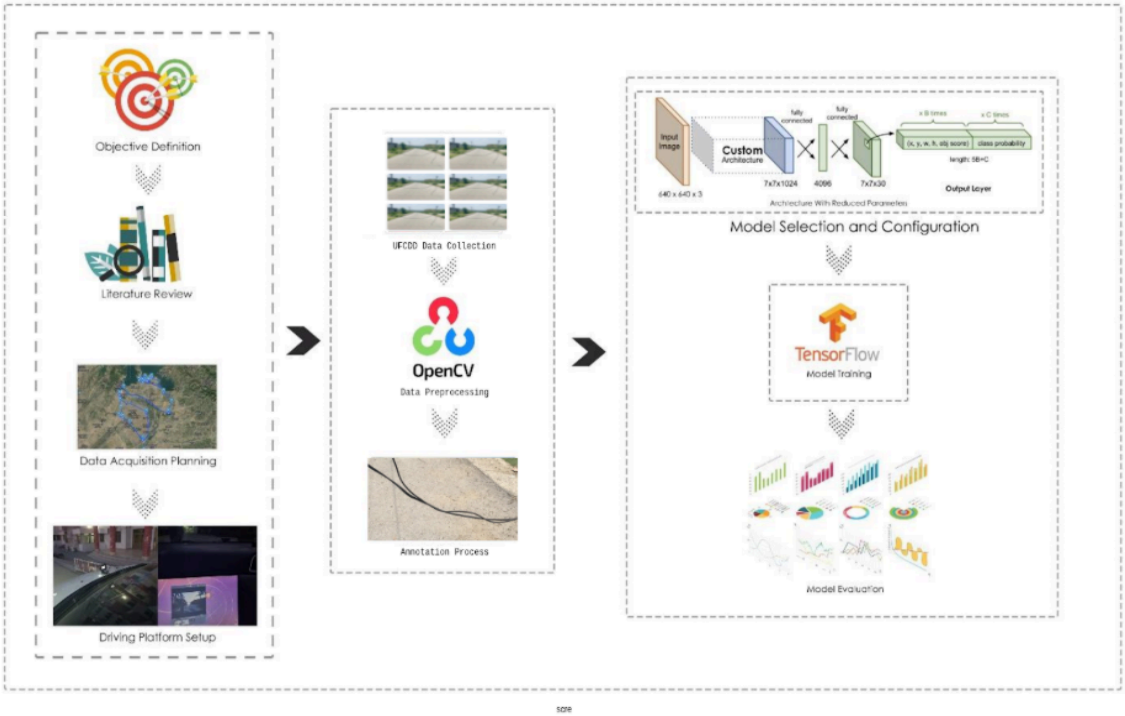

Figure 2 illustrates the complete system workflow from data collection to prediction.

RESULTS AND DISCUSSION

The experimental phase evaluates the customized CNN-based YOLOv5 model trained on the Urban Fallen Cable Detection Dataset (UFCDD). The goal is to validate real-time hazard detection performance and localization precision in complex urban scenes.

The model was trained for 300 epochs over 72 hours using an NVIDIA RTX 2080Ti GPU, Intel Core i9 9900K CPU, and 32 GB RAM [60]. The dataset split was 70% training, 20% validation, and 10% testing. Model scaling parameters were set to width 0.50 and depth 0.33 to balance accuracy and computational efficiency.

- Performance was evaluated using Precision, Recall, F1-score, mAP@50, and mAP@50–95.
- mAP@50 measures detection confidence with 50% IoU overlap.
- mAP@50–95 evaluates strict localization across IoU thresholds from 0.5 to 0.95.

Dataset Characteristics

UFCDD contains approximately 4,000 Safe cable instances and 2,800 Unsafe cable instances. The distribution is shown in Figure 3.

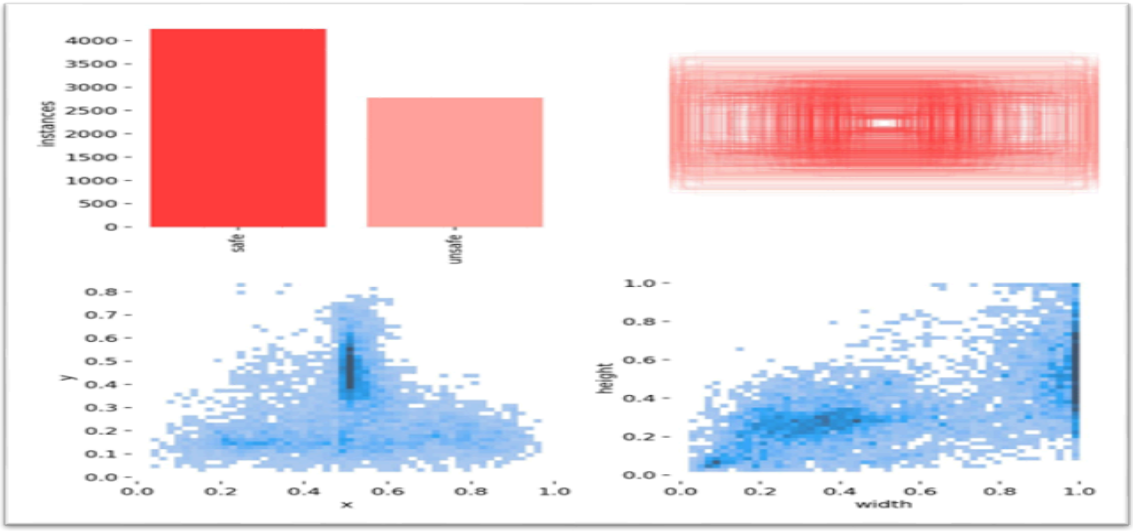

Figure 3: UFCDD Instance Distribution and Geometric Metadata

Most hazards appear in the central vertical region of images, which corresponds to the vehicle path. Bounding box analysis confirms that most objects occupy less than 10% of the image area. This validates the task's small-object nature and justifies the specialized detection configuration.

The experimental phase evaluates the customized CNN-based YOLOv5 model trained on the Urban Fallen Cable Detection Dataset (UFCDD). The goal is to validate real-time hazard detection performance and localization precision in complex urban scenes.

Table 2: summarizes model performance on the UFCDD test set.

Metric	Value	Interpretation
mAP@50	96%	High classification confidence
mAP@50–95	76%	Moderate localization precision
GFLOPs	15.8	Real-time efficiency
Parameters	7M	Edge deployment suitability

The model achieves strong classification accuracy. Localization performance is lower under strict IoU evaluation.

Classification Performance

Figure 4a Recall-Confidence curve demonstrates that there is recall of more than 0.90 in the unsafe class with a broad confidence interval. This makes sure that the hazards that are critical are not overlooked.

Figure 4b shows that the Precision-Recall Curve of Unsafe cables has mAP@0.5 of 0.983 and 0.965, respectively. This ascertains high detection ability.

Figure 4c Precision-Confidence curve indicates almost perfect precision at high levels of confidence. The level of confidence of 0.989 gives a precision of 1.00. Detected cases with high confidence are virtually correct.

Figure 4d F1-Confidence curve is at its highest point of 0.93 at a 0.407 confidence level. This is the best precision and recall tradeoff.

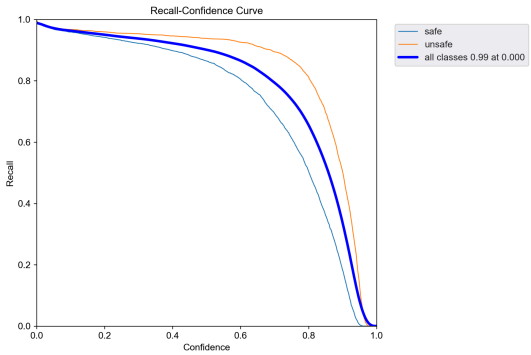

Figure 4a

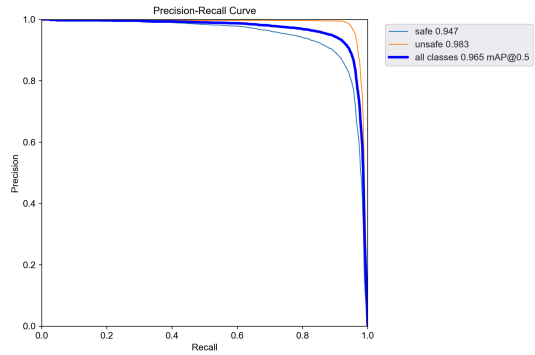

Figure 4b

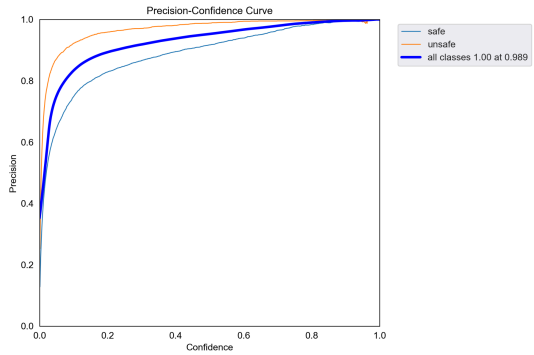

Figure 4c

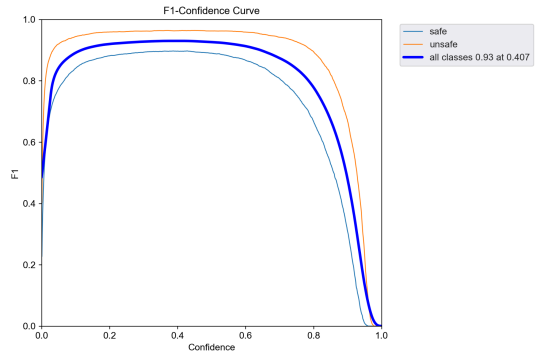

Figure 4d

Figure 4: Validation metrics

Performance is also validated by the confusion matrix in Figure 5. Unsafe cables have a true Positive rate of 95%, and Safe cables have a true Positive rate of 94%. False negatives are still between 4 and 6. This means that thin cable structures can be well recovered even in disorganized backgrounds.

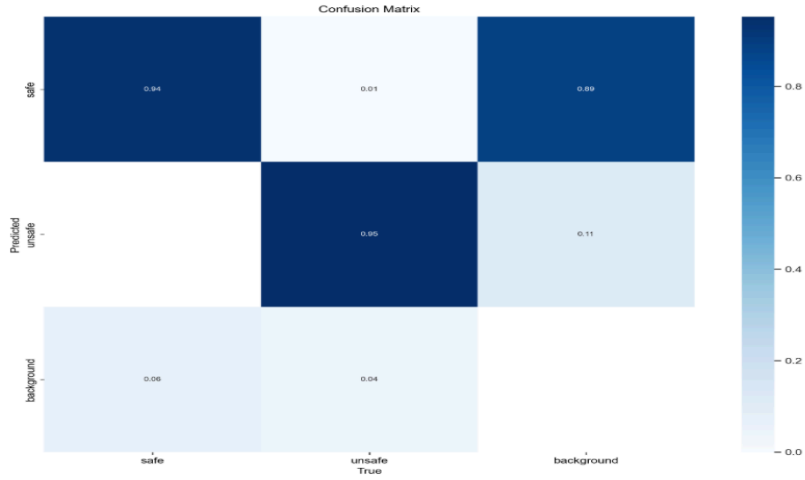

Figure 5: Confusion Matrix for Cable Classification

The success of the classification performance is caused by two factors:
Dataset fidelity: UFCDD models real cable geometry and variations in lighting.

Specialized feature learning: YOLOv5 is able to differentiate between cables and shadows as well as road cracks.

Localization Deficit

Regardless of high detection confidence, stricter performance of localization is 76% mAP5095. This indicates a 20 percent difference between detection and accurate placement of the boundaries. Figure 4 mAP evolution indicates that mAP 0.5 is converging to 0.965. However, mAP@0.5:0.95 plateaus near 0.78. This implies a geometrical constraint. Thin cables may only be 1-3 pixels long [28]. Such curvilinear structures do not fit well with rectangular bounding boxes [34]. It is geometrically hard to achieve a high IoU on very thin objects [33]. This justifies the Geometric Ambiguity Problem outlined in previous research [27]. Detection is reliable. Defining boundaries according to accuracy is still difficult.

Training Convergence and Dynamics.

Figure 6 represents the learning rate schedule. There is a warm-up period, which stabilizes gradients. A phase of gradual decay makes it convergent.

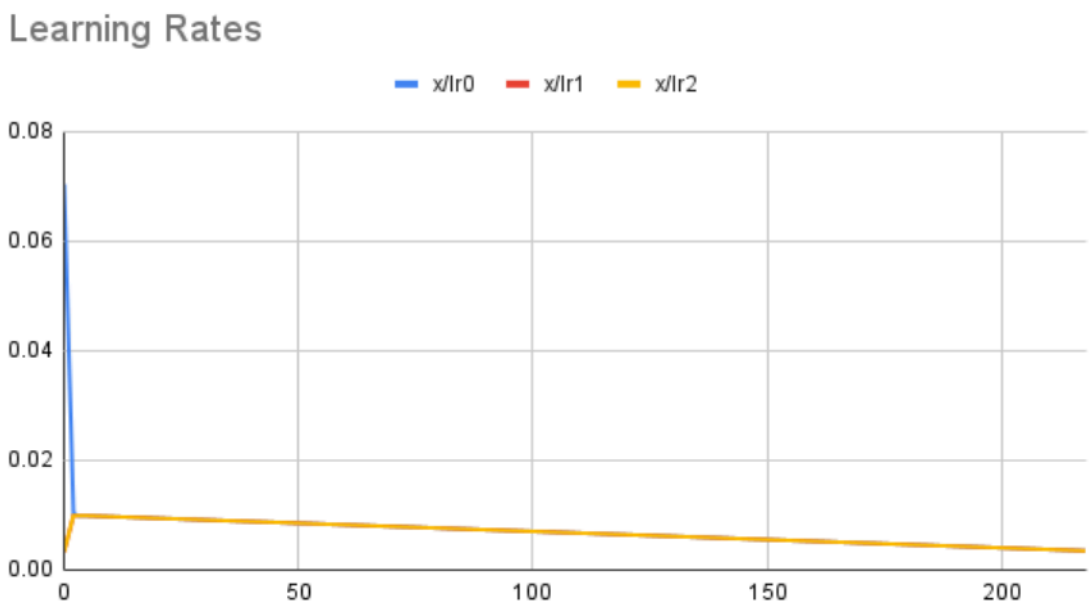

Figure 6: Learning Rate Schedules

Training and validation loss curves are shown in Figure 4. The losses of bounding box and objectness decrease gradually. The method of classification approaches zero. There is no deviation between training and validation curves. The problem of overfitting is not visible. These tendencies confirm constant optimization across 300 epochs.

Comparative Robustness

The trends of performance are in line with other data sets that are geometry sensitive. Objects that are structured like vehicles are normally highly localized. The performance of strict IoU in irregular objects is lower [64].

The mAP 50-95 of 76% on UFCDD corresponds to the thinness and curvilinear nature of cables. It is more difficult than inflexible vehicle detection and more stable than highly deformable human poses.

This limitation should be overcome in the future by research into segmentation-based approaches and specialized loss functions.

Real-Time Feasibility

The customized YOLOv5 architecture contains 157 layers and 7 million parameters. Calculation cost 15.8 GFLOPs.

This proves to be appropriate for deploying edges in autonomous vehicles. Operation at low latency is possible.

The framework illustrates that systems of AV perception can be modified to fit infrastructure hazards in a region [9], [10].

Table 3: Research questions validations

Research Question	Result
RQ1 Dataset Fidelity	Validated
RQ2 Model Suitability	Validated
RQ3 Localization Precision	Partially Validated
RQ4 Scalability Impact	Validated

UFCDD effectively manages the lack of hazards of HSLF cable as standard benchmarks. The trained YOLOv5 model is highly accurate in detection and can work in real time. The major technical problem is the precision of localization.

CONCLUSIONS

In this paper, a new visual standard of identifying fallen electric cables in inner-city urban areas in Pakistan was proposed. Fallen cables are not like traditional traffic objects because they pose special detection problems, as they have a thin structure, low contrast, and do not stay in the same orientation. Baseline experiments show how ineffective the current object detection frameworks are when used on these rare and infrastructure-induced hazards.

This work presents a baseline assessment and specific data set that can serve as a foundation of future studies in the area of safety-critical perception systems on autonomous cars and intelligent mobility solutions in developing areas.

Further areas of research will be:

1. A combination of segmentation-based models to enhance the localization of thin objects.
2. Research on transformer-based systems of detection to enable better contextual comprehension.
3. Depth and thermal sensor multi-modal fusion.
4. Field tests and real-life implementation of smart mobility pilot zones in Pakistan.
5. Plans for domain generalization in case of transferability of the strategies to other developing countries.

AUTHOR CONTRIBUTIONS

The authors confirm contribution to the paper as follows: study conception and design: X. Author, Y. Author; data collection: Y. Author; analysis and interpretation of results: X. Author, Y. Author. Z. Author; draft manuscript preparation: Y. Author. Z. Author. All authors reviewed the results and approved the final version of the manuscript.

REFERENCES

1. Patil, Ashok Kumar, Bhargav Punugupati, Himanshi Gupta, Niranjan S. Mayur, Srivatsa Ramesh, and Prasad B. Honnavalli. "Building the Future of Transportation: A Comprehensive Survey on AV Perception, Localization, and Mapping." Sensors 25, no. 7 (2025): 2004.

2. Gan, Lu, Wenbo Chu, Guofa Li, Xiaolin Tang, and Keqiang Li. "Large models for intelligent transportation systems and autonomous vehicles: A survey." Advanced Engineering Informatics 62 (2024): 102786.

3. Gherardini, Luca, and Giacomo Cabri. "The impact of autonomous vehicles on safety, economy, society, and environment." World Electric Vehicle Journal 15, no. 12 (2024): 579.

4. Valdovinos, Samantha, Claire Barbrack, and Nicole Podest. "Are Autonomous Vehicles Really "Autonomous?": The Importance of Human Factors in AV Safety." In Proceedings of the Human Factors and Ergonomics Society Annual Meeting, vol. 67, no. 1, pp. 1602-1605. Sage CA: Los Angeles, CA: SAGE Publications, 2023.

5. Valverde, Miguel, Alexandra Moutinho, and João-Vitor Zacchi. "A Survey of Deep Learning-Based 3D Object Detection Methods for Autonomous Driving Across Different Sensor Modalities." Sensors (Basel, Switzerland) 25, no. 17 (2025): 5264.

6. Yan, Huan, and Yong Li. "Generative AI for Intelligent Transportation Systems: Road Transportation Perspective." ACM Computing Surveys (2025).

7. Contreras, Marcelo, Aayush Jain, Neel P. Bhatt, Arunava Banerjee, and Ehsan Hashemi. "A survey on 3D object detection in real time for autonomous driving." Frontiers in Robotics and AI 11 (2024): 1212070.

8. Chen, Zhige, Zhigang Zhang, Qizheng Su, Kai Yang, Yandong Wu, Lei He, and Xiaolin Tang. "Object detection for autonomous vehicles under adverse weather conditions." Expert Systems with Applications (2025): 128994.

9. Yan, Guangjian, Chaoyang Li, Guoqing Zhou, Wuming Zhang, and Xiaowen Li. "Automatic extraction of power lines from aerial images." IEEE Geoscience and remote sensing letters 4, no. 3 (2007): 387-391.

10. Liu, Chenxi. "Situation-aware Customized Machine Intelligence for Transportation Safety, Equity, and Resilience." PhD diss., University of Washington, 2024.

11. Hassija, Vikas, Balamurugan Palanisamy, Arpita Chatterjee, Arpita Mandal, Debanshi Chakraborty, Amit Pandey, G. S. S. Chalapathi, and Dhruv Kumar. "Transformers for Vision: A Survey on Innovative Methods for Computer Vision." IEEE Access (2025).

12. Bozkir, Efe, Süleyman Özdel, Mengdi Wang, Brendan David-John, Hong Gao, Kevin Butler, Eakta Jain, and Enkelejda Kasneci. "Eye-tracked virtual reality: A comprehensive survey on methods and privacy challenges." arXiv preprint arXiv:2305.14080 (2023).

13. Garcia, F., Cerri, P., & Broggi, A. (2017). "Sensor Fusion for Autonomous Vehicles in Urban Environments." IEEE Robotics and Automation Letters, 3(4), 3214-3221. DOI: 10.1109/LRA.2018.2849491.

14. Johnson, R.E., & Thompson, M.K. (2022). "Enhancing Autonomous Driving Algorithms through Visualization of Complex Datasets." Advanced Computing Research, 7(2), 344-360.

15. Gupta, S., & Kumar, P. (2021). Enhancing Low-Light Videos for Autonomous Driving Systems: A Deep Learning Approach. Pattern Recognition Letters.

16. Cordts, M., Omran, M., Ramos, S., Rehfeld, T., Enzweiler, M., Benenson, R., Franke, U., Roth, S., & Schiele, B. (2016). "The Cityscapes Dataset for Semantic Urban Scene Understanding." Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 3213-3223.

17. Patel, R., & Sengupta, R. (2020). "Autonomous Vehicles: Navigating the Regulatory and Infrastructure Challenges in Developing Nations." Journal of Urban Planning and Development, 146(1), 05019019.

18. Faisal, Md Ahasan Atick, Imene Mecheter, Yazan Qiblawey, Javier Hernandez Fernandez, Muhammad EH Chowdhury, and Serkan Kiranyaz. "Deep learning in automated power line inspection: A review." Applied Energy 385 (2025): 125507.

19. Omrani, Farzane, Hossein Etemadfard, and Rouzbeh Shad. "Assessment of aviation accident datasets in severity prediction through machine learning." Journal of Air Transport Management 115 (2024): 102531.

20. Mohammed, Anaam, Miaad K. Alkhudhairy, and Ali A. Aldhalemi. "Fatal, Non-Fatal Injuries, and Deaths Resulting from Traffic Accidents for the Years 2018-2023 in Iraq." Medical Science Journal for Advance Research 6, no. 1 (2025).

21. Karami, Omid. "3D object detection via LiDAR-camera fusion in autonomous driving." (2024).

22. Dalal, Navneet, and Bill Triggs. "Histograms of oriented gradients for human detection." In 2005 IEEE computer society conference on computer vision and pattern recognition (CVPR'05), vol. 1, pp. 886-893. Ieee, 2005.

23. Ross, T-YLPG, and G. K. H. P. Dollár. "Focal loss for dense object detection." In proceedings of the IEEE conference on computer vision and pattern recognition, pp. 2980-2988. IEEE, 2017.

24. Janai, Joel, Fatma Güney, Aseem Behl, and Andreas Geiger. "Computer vision for autonomous vehicles: Problems, datasets and state of the art." Foundations and trends® in computer graphics and vision 12, no. 1–3 (2020): 1-308.

25. Grigorescu, Sorin, Bogdan Trasnea, Tiberiu Cocias, and Gigel Macesanu. "A survey of deep learning techniques for autonomous driving." Journal of field robotics 37, no. 3 (2020): 362-386.

26. Zhu, Hufei, Yonghui Huang, Ying Xu, Jianhong Zhou, Fuqin Deng, and Yikui Zhai. "Unmanned aerial vehicle (UAV) object detection algorithm based on keypoints representation and rotated distance-IoU loss." Journal of Real-Time Image Processing 21, no. 2 (2024): 58.

27. Liu, Songwen, Hao He, Zhichao Zhang, and Yatong Zhou. "LI-YOLO: An Object Detection Algorithm for UAV Aerial Images in Low-Illumination Scenes." Drones 8, no. 11 (2024): 653.

28. Zhao, Zhong-Qiu, Peng Zheng, Shou-tao Xu, and Xindong Wu. "Object detection with deep learning: A review." IEEE transactions on neural networks and learning systems 30, no. 11 (2019): 3212-3232.

29. Liu, Wei, Dragomir Anguelov, Dumitru Erhan, Christian Szegedy, Scott Reed, Cheng-Yang Fu, and Alexander C. Berg. "Ssd: Single shot multibox detector." In European conference on computer vision, pp. 21-37. Cham: Springer International Publishing, 2016.

30. Sandler, Mark, Andrew Howard, Menglong Zhu, Andrey Zhmoginov, and Liang-Chieh Chen. "Mobilenetv2: Inverted residuals and linear bottlenecks." In Proceedings of the IEEE conference on computer vision and pattern recognition, pp. 4510-4520. 2018.

31. Tremblay, Jonathan, Aayush Prakash, David Acuna, Mark Brophy, Varun Jampani, Cem Anil, Thang To, Eric Cameracci, Shaad Boochoon, and Stan Birchfield. "Training deep networks with synthetic data: Bridging the reality gap by domain randomization." In Proceedings of the IEEE conference on computer vision and pattern recognition workshops, pp. 969-977. 2018.

32. Wigness, Maggie, Sungmin Eum, John G. Rogers, David Han, and Heesung Kwon. "A rugd dataset for autonomous navigation and visual perception in unstructured outdoor environments." In 2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 5000-5007. IEEE, 2019.

33. Ison, David C. "Comparative analysis of accident and non-accident pilots." Journal of Aviation Technology and Engineering 4, no. 2 (2015): 20.

34. Huang, Chenyu. "Analysis of part 135 aircraft accidents to facilitate flight data monitoring." International Journal of Aviation, Aeronautics, and Aerospace 7, no. 1 (2020).

35. Caesar, Holger, Varun Bankiti, Alex H. Lang, Sourabh Vora, Venice Erin Liong, Qiang Xu, Anush Krishnan, Yu Pan, Giancarlo Baldan, and Oscar Beijbom. "nuscenes: A multimodal dataset for autonomous driving." In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 11621-11631. 2020.

36. Azad, Reza, Moein Heidary, Kadir Yilmaz, Michael Hüttemann, Sanaz Karimijafarbigloo, Yuli Wu, Anke Schmeink, and Dorit Merhof. "Loss functions in the era of semantic segmentation: A survey and outlook." arXiv preprint arXiv:2312.05391 (2023).

37. Bouguettaya, Abdelmalek, Ahmed Kechıda, and Amine Mohammed Taberkıt. "A survey on lightweight CNN-based object detection algorithms for platforms with limited computational resources." International Journal of Informatics and Applied Mathematics 2, no. 2 (2019): 28-44.

38. Choe, Tae Eun, Jane Wu, Xiaolin Lin, Karen Kwon, and Minwoo Park. "Hazardnet: Road debris detection by augmentation of synthetic models." In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 161-171. 2023.

39. Chong, Qianpeng, Mengying Ni, Jianjun Huang, Guangyi Wei, Ziyi Li, and Jindong Xu. "Let the loss impartial: a hierarchical unbiased loss for small object segmentation in high-resolution remote sensing images." European Journal of Remote Sensing 56, no. 1 (2023): 2254473.

40. Damodaran, Satheeswari, Leninisha Shanmugam, and N. M. Jothi Swaroopan. "Overhead power line detection from aerial images using segmentation approaches." Automatika: časopis za automatiku, mjerenje, elektroniku, računarstvo i komunikacije 65, no. 1 (2024): 261-288.

41. Wigness, Maggie, Sungmin Eum, John G. Rogers, David Han, and Heesung Kwon. "A rugd dataset for autonomous navigation and visual perception in unstructured outdoor environments." In 2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 5000-5007. IEEE, 2019.

42. Qiao, Wenting, Biao Ma, Qiangwei Liu, Xiaoguang Wu, and Gang Li. "Computer vision-based bridge damage detection using deep convolutional networks with expectation maximum attention module." Sensors 21, no. 3 (2021): 824.

43. Gao, Xiaofei. Deep Learning-Based Road Spill Detection Technology. No. 2025-01-7130. SAE Technical Paper, 2025.

44. Gebresenbet, Girma. "Prospects for Implementation of Autonomous Vehicles and Associated Infrastructure in Developing Countries." (2024).

45. Kolbeinsson, Benedikt, and Krystian Mikolajczyk. "UCorr: Wire Detection and Depth Estimation for Autonomous Drones." In International Conference on Robotics, Computer Vision and Intelligent Systems, pp. 179-192. Cham: Springer Nature Switzerland, 2024.

46. Ren, Yuemei, and Lei Li. "Line segment detection algorithm in image extraction improvement study." Journal of Measurements in Engineering 12, no. 1 (2024): 199-213.

47. Boroujeni, Sayed Pedram Haeri, Niloufar Mehrabi, Hazim Alzorgan, Ahmad Sarlak, Mahlagha Fazeli, and Abolfazl Razi. "All You Need for Object Detection: From Pixels, Points, and Prompts to Next-Gen Fusion and Multimodal LLMs/VLMs in Autonomous Vehicles." arXiv preprint arXiv:2510.26641 (2025).

48. Liew, Jun Hao, Scott Cohen, Brian Price, Long Mai, and Jiashi Feng. "Deep interactive thin object selection." In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, pp. 305-314. 2021.

49. Huang, Li, Cheng Chen, Juntong Yun, Ying Sun, Jinrong Tian, Zhiqiang Hao, Hui Yu, and Hongjie Ma. "Multi-scale feature fusion convolutional neural network for indoor small target detection." Frontiers in Neurorobotics 16 (2022): 881021.

50. Madaan, Ratnesh, Daniel Maturana, and Sebastian Scherer. "Wire detection using synthetic data and dilated convolutional networks for unmanned aerial vehicles." In 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 3487-3494. IEEE, 2017.

51. Moltz, Jan Hendrik, Annika Hänsch, Bianca Lassen-Schmidt, Benjamin Haas, A. Genghi, J. Schreier, Tomasz Morgas, and Jan Klein. "Learning a loss function for segmentation: A feasibility study." In 2020 IEEE 17th International Symposium on Biomedical Imaging (ISBI), pp. 357-360. IEEE, 2020.

52. O'Mahony, Niall, Sean Campbell, Anderson Carvalho, Suman Harapanahalli, Gustavo Velasco Hernandez, Lenka Krpalkova, Daniel Riordan, and Joseph Walsh. "Deep learning vs. traditional computer vision." In Science and information conference, pp. 128-144. Cham: Springer International Publishing, 2019.

53. Ogundairo, Obaloluwa, Marty Gregg, Kristina Holsgrove, Ying Li, and Mark Allan Kaplan. "A Deep Dive into Boundary-Aware Loss Functions for Fine-Grained Semantic Segmentation." (2024).

54. Woo, Sanghyun, Shoubhik Debnath, Ronghang Hu, Xinlei Chen, Zhuang Liu, In So Kweon, and Saining Xie. "Convnext v2: Co-designing and scaling convnets with masked autoencoders." In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 16133-16142. 2023.

55. Sang, Shengtian, Yuyin Zhou, Md Tauhidul Islam, and Lei Xing. "Small-object sensitive segmentation using across feature map attention." IEEE transactions on pattern analysis and machine intelligence 45, no. 5 (2022): 6289-6306.

56. Chaturvedee, Adarsh, Raghad Al-Shabandar, and Ammar H. Mohammed. "Small Object Detection in Autonomous Cars Using a Deep Learning." International Journal of Data Science and Advanced Analytics 6, no. 6 (2024): 307-314.

57. Sun, X., Y. Hou, and P. W. T. Pong. "Underground Power Cable Detection and Inspection Technology Based on Magnetic Field Sensing and Source Reconstruction." In International Symposium on Advanced Magnetic Materials and Applications, ISAMMA 2013. 2013.

58. Wang, Kaige, Tianming Wang, Jianchuang Qu, Huatao Jiang, Qing Li, and Lin Chang. "An end-to-end cascaded image deraining and object detection neural network." IEEE Robotics and Automation Letters 7, no. 4 (2022): 9541-9548.

59. Wang, Xiaoyan, and Huizong Li. "Investigation of Deep Learning Based Semantic Segmentation Models for Autonomous Vehicles." International Journal of Advanced Computer Science & Applications 14, no. 11 (2023).

60. Yi, Wenxin, Zhang Zhang, and Liang Chang. "M4MLF-YOLO: A Lightweight Semantic Segmentation Framework for Spacecraft Component Recognition." Remote Sensing 17, no. 18 (2025): 3144.

61. Elharrouss, Omar, Yasir Mahmood, Yassine Bechqito, Mohamed Adel Serhani, Elarbi Badidi, Jamal Riffi, and Hamid Tairi. "Loss functions in deep learning: a comprehensive review." arXiv e-prints (2025): arXiv-2504.

62. Liu, Jun, Zhenglun Kong, Pu Zhao, Weihao Zeng, Hao Tang, Xuan Shen, Changdi Yang et al. "Tsla: A task-specific learning adaptation for semantic segmentation on autonomous vehicles platform." IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (2024).

63. Zhang, Yijie, Zhong Liu, Hui Nie, Yuechen Yu, and Guannan Chang. "Vehicle target detection based on ConvNext-improved YoloV5." In International Conference on Image Processing and Deep Learning (IPDL 2025), vol. 13707, pp. 163-169. SPIE, 2025.

64. Zhou, Chen, Jiaolong Yang, Chunshui Zhao, and Gang Hua. "Fast, accurate thin-structure obstacle detection for autonomous mobile robots." In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops, pp. 1-10. 2017.

65. Gwizdała, Jakub, Doruk Oner, Soumava Kumar Roy, Mian Akbar Shah, Ad Eberhard, Ivan Egorov, Philipp Krüsi, Grigory Yakushev, and Pascal Fua. "Vision-based power line cables and pylons detection for low flying aircraft." Machine Vision and Applications 36, no. 2 (2025): 46.

Appendices/Supplemental Material

The TRR allows appendices/supplemental material to be included with submissions. For review purposes, please place them at the end of your paper after the references. All supplemental material must be referenced in the manuscript text as well – the preferred labels for them are Table S1, Figure S2, etc.

Great question — adding a few **well-placed equations** will significantly strengthen your paper (especially for an Elsevier CSL submission). You don't need many — **2–3 core equations** are enough if they are meaningful and tied to your methodology.

Here are the **best 3 equations**, with **where to place them** and **1–2 line explanations**.

✅ 1. LoRA Update Equation

📍 Where to add

- **Chapter 4 (Methodology)**
- **Section 4.1: Architecture of the LADI Framework**

📌 Equation

W' = W + BA

✏️ Explanation (add below equation)

Here, (W) represents the original pre-trained weight matrix, while (B) and (A) are low-rank matrices introduced by LoRA. This formulation enables efficient fine-tuning by updating only a small subset of parameters while keeping the base model frozen [24].

✅ 2. Adversarial Debiasing Objective (Minimax Loss)

📍 Where to add

- **Chapter 4 (Methodology)**
- **Section 4.2: Training Strategy and Optimization**

📌 Equation

\min _ {\theta} \max_{\phi} ; \mathcal{L}*{task}(\theta) - \lambda \mathcal{L}*{adv}(\theta, \phi)

✏️ Explanation

The model parameters (\theta) are optimized to minimize task loss while maximizing adversarial loss, ensuring that sensitive attributes cannot be predicted from learned representations. The hyperparameter (\lambda) controls the trade-off between performance and fairness [26].

✅ 3. ICAT Score (Bias–Performance Trade-off)

📍 Where to add

- **Chapter 5 (Results)**
- **Section 5.1: Experimental Setup and Evaluation Metrics**

📌 Equation

ICAT = LM \times (1 - |Bias|)

✏️ Explanation

ICAT measures the balance between language modeling performance (LM) and bias, penalizing models that achieve high accuracy at the cost of fairness. Higher ICAT values indicate better overall trade-offs [19].

🎯 Why these 3 are perfect (important insight)

- **LoRA equation** → shows *technical novelty / efficiency*
- **Adversarial loss** → shows *core debiasing mechanism*
- **ICAT** → shows *evaluation rigor*

👉 Together, they cover:

- **Model design**
- **Training objective**
- **Evaluation**

This is exactly what reviewers look for.

⚠️ Important tip (very important for acceptance)

Do **NOT** overload with equations. CSL papers prefer:

- Few equations
- But **highly meaningful and referenced in text**

🚀 If you want next level polish

I can:

- Insert these equations **directly into your chapters (perfect placement + formatting)**
- Convert everything into **Elsevier LaTeX (with equation numbering)**
- Or suggest **one diagram that matches each equation (very strong for reviewers)**